

Charlie Chavez Meelboom

cchavezmeelboo7852@sdsu.edu

(949) 439-4010

Education

San Diego State University

B.S. Mechanical Engineering | Minor: Mathematics | GPA: 3.81

Coursework: Mechanics of Materials, Fluid Mechanics, MATLAB, Physics 3 (Thermodynamics & Optics), Mechatronics

San Diego, CA

Expected May 2028

Leadership

Aztec Electric Racing (Formula SAE Electric)

Operations Lead/Powertrain Engineer

San Diego, CA

Jun 2025 – Present

- Built AER's first business/operations department from the ground up; recruited 20+ members through the Fowler College of Business and business club partnerships.
- Ran weekly execution meetings, and helped secure over \$120,000+ in funding through alumni events and active sponsorship outreach.
- Led cost-report effort twice, leading sub team coordination of GD&T drawing collection and revisions to deliver 700+ page FSAE compliant report in under four months.
- Owned mechanical design of safety-critical housings (RML, Fill Can) in SolidWorks; iterated Bambu X1 Carbon prototypes to validate fit, assembly, and sealing, delivering a fully waterproof RML housing.
- Supported manufacturing through rear-wing carbon-fiber wet layups, waterjet fabrication for tabs and rotors and 3D printing of housings.
- Supported design through Reset Latch PCB documentation; GD&T component powertrain drawings, test-data collection for design validation, and powertrain tab-thickness structural analysis

SDSU College of Engineering Student Council (CESC)

VP of Finance (Elected)

San Diego, CA

Jun 2025 – Present

- Developed CESC's first unified Excel system for budgeting and spending, boosting transparency across all accounts and reducing reporting time.
- Rewrote reimbursement documentation into single page process used by all engineering organizations at SDSU decreasing reimbursement time by over a week.
- Increased university funding by \$7,000 through direct lobbying with the Head of Financial Affairs; launched the first-ever Order of the Ring fundraiser, raising an additional \$600+.

SDSU Dynamic Systems and Intelligent Machines (DSIM) Lab

Undergraduate Researcher (AUV acoustic-localization project)

San Diego, CA

Aug 2025 – Present

- Improved test repeatability and data quality by documenting pool-test procedures and results. Designed hydrophone mounts and tight-tolerance housings; integrated the full sensor/servo stack.
- Developed the data pipeline using ROS2 nodes and Arduino IDE that relays sensor angle outputs from Raspberry Pi and DAC HAT for processing within an extremum seeking control framework.

KDG Industrial

Sales & Warehouse Clerk (Seasonal)

San Diego, CA

Sep 2022 – Jan 2025

- Reduced quoting times by 5 days for a high-volume international customer account through tighter coordination with technical and warehouse teams and weekly task meetings with team.
- Implemented HubSpot CRM workflows and training documentation that became the company's primary deal-flow platform; oversaw material reception, inventory checks, and local deliveries, and supported product exports to Mexico.

Projects

Self-Sorting Waste Bin Prototype (Innovate 4 SDSU Hackathon)

3rd Place ("Most Impactful")

- Built a live demo that classifies compost/landfill/recycling from real-time sensor readings and actuates sorting behavior using an Arduino-based sensor setup. Developed a KNN classifier achieving 74% accuracy; led the final pitch and demo.

Skills

Proficient in Microsoft Office, Arduino IDE, Python, Git, 3D Printing, Soldering/Wiring, Waterjetting, Test Documentation1, ANSYS Mechanical, MATLAB, ROS2, Linux, Manufacturing, Management

Languages: English (fluent), Spanish (fluent)